

Masters in Mechanical Engineering

Static Analysis of a Wing Box

Aeronautical Structural Design and Analysis I

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Introduction

The current report depicts a project carried out in the Aeronautical Structural Design and Analysis I course, as part of the Masters in Mechanical Engineering at FEUP. The goal of the said project was to perform the static structural analysis of a wing box.

The project can be divided into two main parts: the development of a finite element model and the computation of the Margins of Safety required for certification purposes.

In terms of report structure, in Chapter 1 the problem is presented, *i. e.*, the geometry, materials and load scenario are defined. Then, Chapter 2 deals with all the aspects of the finite element modelling, including the post processing of the results. The finite element model was developed using the software ABAQUS.

Following the numerical simulations, the computation of the Margins of Safety takes place in Chapter 3. This analysis was conducted for the most common structure failure modes, following NACA Report 3781, 1928 and 2661 and Lockheed Report No. 18370.

Then, overall analysis and comments to the obtained results are presented in Chapter 4. Finally, the project conclusions are presented in Chapter 5

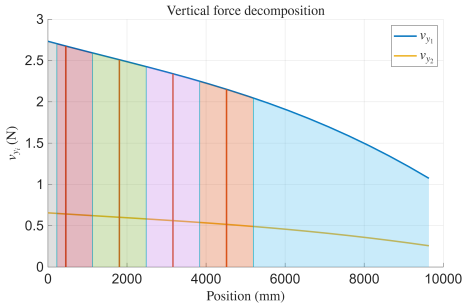


Figure 30: Integration limits for the aerodynamic load calculation

Table 12: Integration limits considered for each rib.

Rib	Inferior Limit [mm]	Rib Position [mm]	Superior Limit [mm]
N1	225.900	451.800	1129.500
N2	1129.500	1807.200	2484.900
N3	2484.900	3162.600	3839.400
N4	3839.400	4518.000	5195.700

The mentioned integration was employed in the same MATLAB code, following the load decomposition, and yielded the following results presented in Table 13.

In turn, the shear loads acting over the tip that is not modelled is estimated by a numerical approximation, given by the Equations 2.9 and 2.10, applied along the points provided in Table 3.

$$V_i = V_{i-1} + \frac{v_i + v_{i-1}}{2}(Y_{i-1} - Y_i) \tag{2.9}$$

$$M_i = M_{i-1} + \frac{V_i + V_{i-1}}{2}(Y_{i-1} - Y_i) \tag{2.10}$$

where v is the distributed shear force and Y the distance from the considered point to

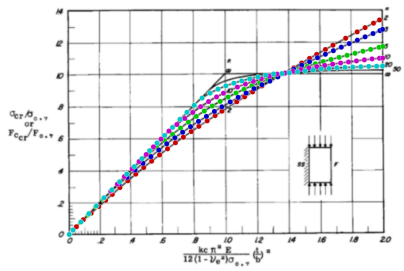


Figure 53: Sets of points gathered by digitizing a graph

The aforementioned subroutine – allowing the main framework to perform data lookups – was also coded specifically for this project. To provide any point data requested, linear interpolations are performed based on the digitized sets.

3.1 Tension

Due to the ductile material characteristics of the aluminum alloy utilized in the wing-box structure, the Von Mises yield criterion was implemented to calculate the tensile margins of safety. The analytical expression governing this calculation is defined in equation 3.1:

$$MS = \frac{F_{tu}}{f_{vm}} - 1 \tag{3.1}$$

Where:

- MS = Margin of Safety;
- F_{tu} = Ultimate tensile strength of the material;
- f_{vm} = Maximum Von Mises stress derived from the finite element model.

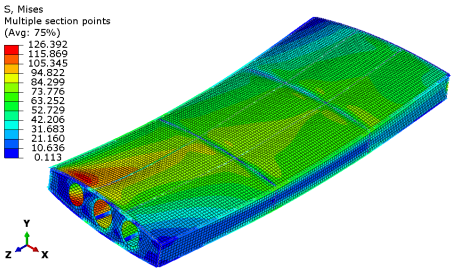


Figure 43: Global equivalent von Mises stress obtained in ABAQUS

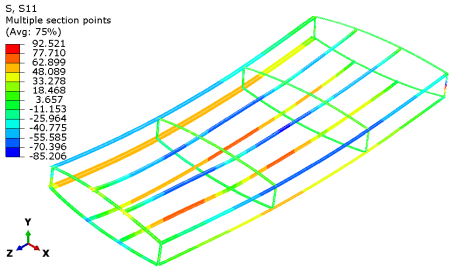


Figure 44: S11 field obtained for the stringers in ABAQUS

Considering the deformed configuration of the structure it complies with the expected for the analysed flight condition: the wing bends upwards with increasing displacement magnitude as we move further away from the root, which is fixed. That is the typical behaviour of a beam-like structure. Additionally, considering the region near the N4 section, the front part exhibits higher displacement magnitude compared to the rear part. This is also expected given the torsional moment provocked by the aerodynamic

The values obtained for the effective geometric and working stress parameters for each iterations are listed in table 31 and 32, for rib N2 and N3 respectively:

Table 31: Iterative loop history for rib N2 effective flange metrics.

Step	f_b [MPa]	w [mm]	w_{min} [mm]	Error
1	-60.278	19.509	15.895	–
2	-65.107	16.863	15.294	0.080110
3	-66.234	15.867	15.164	0.017309
4	-66.618	15.426	15.120	0.005809
5	-66.778	15.216	15.102	0.002399
6	-66.852	15.112	15.094	0.001099
7	-66.887	15.060	15.090	0.000531

Table 32: Iterative loop history for rib N3 effective flange metrics.

Step	f_b [MPa]	w [mm]	w_{min} [mm]	Error
1	-164.583	12.783	9.620	–
2	-187.058	10.292	9.023	0.136558
3	-187.612	9.532	9.010	0.002965
4	-187.075	9.233	9.023	0.002866
5	-186.724	9.103	9.031	0.001876
6	-186.542	9.044	9.036	0.000973

The simplification of the structural interactions within the wing-box model allows for a more isolated evaluation of the localized stress fields. It is assumed in this work, that the compression and shear forces are the sole forces operating on the rib web. This fundamental modeling assumption enables the execution of the iterative process to compute the work stress (f_b) and the corresponding effective flange width (w) without needing to also verify local web buckling. However, in a broader application, this verification constitutes an integral part of the overarching iterative loop itself and must be explicitly considered within the structural sizing routine, as shown in figure 59.